

Benchmark results compiled with -O2: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	167.311	152.908	1286.105	4.724	150.458	288.189	262.314	88.087	191.114	1153.158	4.743	4.665	–	–
countdown.deep.10000	1282.936	601.921	2296.875	2718.059	151.237	284.527	166.969	170.245	196.058	3552.623	4.768	4.773	–	–
catch.shallow.10000	277.635	36.096	1155.68	20.997	1081.834	379308.7	252.514	75.337	860.239	852.075	21.131	20.948	–	–
catch.deep.10000	3244.614	34.042	6705.012	1695.027	1093.28	393773.4	179.119	241.244	854.595	862.236	21.03	20.927	–	–
local.shallow.10000	948.476	466.912	2202.869	4.902	553.462	407054.4	201.201	138.647	792.531	799.685	4.936	4.899	–	–
local.deep.10000	3867.702	464.343	5190.609	4.903	552.71	393756.6	206.628	215.995	800.941	804.413	4.967	4.909	–	–
nondet.shallow.32	1992.041	3272.809	7807.906	245.563	–	25059.088	777.825	911.303	1745.776	1785.469	227.38	247.595	366.988	357.558
nondet.deep.32	5417.919	5784.613	7883.922	2470.542	–	51122.9	788.941	629.105	5744.378	5737.953	248.557	244.059	64318.575	64349.9

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	3586.8%	3278.1%	27571.7%	101.3%	3225.5%	6178.2%	5623.5%	1888.4%	4097.1%	24721.5%	101.7%	100.0%	–	–
countdown.deep.10000	26906.8%	12624.0%	48171.9%	57005.3%	3171.9%	5967.3%	3501.8%	3570.5%	4111.9%	74508.5%	100.0%	100.1%	–	–
catch.shallow.10000	1325.4%	172.3%	5516.9%	100.2%	5164.4%	1810722.2%	1205.4%	359.6%	4106.6%	4067.6%	100.9%	100.0%	–	–
catch.deep.10000	15504.3%	162.7%	32039.7%	8099.6%	5224.2%	1881633.1%	855.9%	1152.8%	4083.7%	4120.2%	100.5%	100.0%	–	–
local.shallow.10000	19359.5%	9530.2%	44963.0%	100.1%	11296.8%	8308441.6%	4106.7%	2829.9%	16176.5%	16322.5%	100.8%	100.0%	–	–
local.deep.10000	78886.4%	9470.8%	105868.7%	100.0%	11273.2%	8031138.5%	4214.4%	4405.5%	16336.1%	16407.0%	101.3%	100.1%	–	–
nondet.shallow.32	876.1%	1439.4%	3433.9%	–	–	11020.8%	342.1%	400.8%	767.8%	785.2%	100.0%	108.9%	161.4%	157.3%
nondet.deep.32	2219.9%	2370.2%	3230.3%	1012.3%	–	20946.9%	323.3%	257.8%	2353.7%	2351.0%	101.8%	100.0%	26353.6%	26366.5%

Benchmark results compiled with -O2: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	1760150	1600143	13361464	24	1121566	2087371	2480237	1040119	3441166	14241103	24	24	–	–
countdown.deep.10000	17760391	10400164	37363344	27121228	1136503	2088076	2000189	3920269	3442685	41443869	24	24	–	–
catch.shallow.10000	4069603	65595	7764880	65585	7846338	413420035	1687789	723950	6081379	6081376	65584	65585	–	–
catch.deep.10000	16870397	66057	55153596	16165071	7864080	414232317	1527781	3367869	6083936	6083936	65585	65584	–	–
local.shallow.10000	7205040	4793284	12630005	24	4326211	409640194	1669445	1029422	5513560	5513563	24	24	–	–
local.deep.10000	16005981	4793723	30232155	24	4341323	410400014	1589470	2549549	5515808	5515811	64	64	–	–
nondet.shallow.32	13454087	15889888	43492494	3606902	–	111260547	1531315	1532743	22646040	22646039	1987325	2003982	3664086	3664078
nondet.deep.32	67560245	45645989	43569825	26810630	–	383225866	1531393	1535143	68636376	68636408	1969093	2010670	469818373	469818393

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	7333958.3%	6667262.5%	55672766.7%	100.0%	4673191.7%	8697379.2%	10334320.8%	4333829.2%	14338191.7%	59337929.2%	100.0%	100.0%	–	–
countdown.deep.10000	74001629.2%	43334016.7%	155680600.0%	113005116.7%	4735429.2%	8700316.7%	8334120.8%	16334454.2%	14344520.8%	172682787.5%	100.0%	100.0%	–	–
catch.shallow.10000	6205.2%	100.0%	11839.6%	100.0%	11963.8%	630367.2%	2573.5%	1103.9%	9272.7%	9272.7%	100.0%	100.0%	–	–
catch.deep.10000	25723.3%	100.7%	84096.1%	24647.9%	11990.9%	631605.8%	2329.5%	5135.2%	9276.6%	9276.6%	100.0%	100.0%	–	–
local.shallow.10000	30021000.0%	19972016.7%	52625020.8%	100.0%	18025879.2%	1706834141.7%	6956020.8%	4289258.3%	22973166.7%	22973179.2%	100.0%	100.0%	–	–
local.deep.10000	66691587.5%	19973845.8%	125967312.5%	100.0%	18088845.8%	1710000058.3%	6622791.7%	10623120.8%	22982533.3%	22982545.8%	266.7%	266.7%	–	–
nondet.shallow.32	878.6%	1037.7%	2840.2%	235.5%	–	7265.7%	100.0%	100.1%	1478.9%	129.8%	129.8%	130.9%	239.3%	239.3%
nondet.deep.32	4411.7%	2980.7%	2845.1%	1750.7%	–	25024.7%	100.0%	100.2%	4482.0%	4482.0%	128.6%	131.3%	30679.2%	30679.2%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	805.176	528.742	1835.438	1813.106	777.799	907.639	1699.402	1692.728	899.805	2249.823	679.874	584.314	–	–
countdown.deep.10000	3001.523	990.785	2928.28	11022.987	746.461	890.927	1761.018	3434.737	1517.808	5386.113	5331.137	6126.294	–	–
catch.shallow.10000	1089.265	310.907	2873.097	3436.816	2141.057	378955	987.952	1117.544	1680.058	1660.194	256.898	227.467	–	–
catch.deep.10000	8221.466	313.546	7115.491	16289.031	2174.713	392794.7	1085.773	3027.867	1666.947	1664.02	4058.761	4144.667	–	–
local.shallow.10000	4568.934	691.208	2720.833	1474.663	1081.31	408641	643.822	674.68	1437.265	1424.01	366.119	341.031	–	–
local.deep.10000	13485.288	695.92	5724.8	6057.175	1096.75	394278.3	652.442	1652.103	1419.297	1428.834	837.337	1680.936	–	–
nondet.shallow.32	7714.484	6096.197	13679.212	2266.739	–	28555.787	4306.573	4376.333	3118.791	3129.478	2605.116	2111.945	7523.228	7035.25
nondet.deep.32	15931.831	8171.219	23461.5	11094.131	–	55240.713	4451.122	9882.138	8047.778	8111.919	7289.166	7371.602	83857.55	84693.35

Benchmark results compiled with -O0: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	152.3%	100.0%	347.1%	342.9%	147.1%	171.7%	321.4%	320.1%	170.2%	425.5%	128.6%	110.5%	–	–
countdown.deep.10000	402.1%	132.7%	392.3%	1476.7%	100.0%	119.4%	235.9%	460.1%	203.3%	721.6%	714.2%	820.7%	–	–
catch.shallow.10000	478.9%	136.7%	1263.1%	1510.9%	941.3%	166597.8%	434.3%	491.3%	738.6%	729.9%	112.9%	100.0%	–	–
catch.deep.10000	2622.1%	100.0%	2269.4%	5195.1%	693.6%	125274.8%	346.3%	965.7%	531.6%	530.7%	1294.5%	1321.9%	–	–
local.shallow.10000	1339.7%	202.7%	797.8%	432.4%	317.1%	119825.2%	188.8%	197.8%	421.4%	417.6%	107.4%	100.0%	–	–
local.deep.10000	2066.9%	106.7%	877.4%	928.4%	168.1%	60431.1%	100.0%	253.2%	217.5%	219.0%	128.3%	257.6%	–	–
nondet.shallow.32	365.3%	288.7%	647.7%	107.3%	–	1352.1%	203.9%	207.2%	147.7%	148.2%	123.4%	100.0%	356.2%	333.1%
nondet.deep.32	357.9%	183.6%	527.1%	249.2%	–	1241.1%	100.0%	222.0%	180.8%	182.2%	163.8%	165.6%	1884.0%	1902.7%

Benchmark results compiled with -O0: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	10640928	6080468	17281847	11920873	5202038	6268897	10801408	11601183	7041725	19361584	4080228	2880408	–	–
countdown.deep.10000	38643716	14881328	41284511	63365121	5218124	6269554	11213136	15371462	9444744	48965259	30882789	30882825	–	–
catch.shallow.10000	7913895	2571779	22351479	13075273	12013479	415979505	5433984	6393799	8161735	8161802	1673341	1593381	–	–
catch.deep.10000	33116273	2572595	57554480	64806045	12032258	416786990	5845703	10164133	8165093	8164582	16926365	18046439	–	–
local.shallow.10000	14981824	5771832	14008645	7688302	6726806	411560571	4008436	4648325	7193783	7193838	2327805	2327860	–	–
local.deep.10000	34184816	5772648	31611405	28490609	6742816	412320940	4020185	6818542	7197133	7196622	5048099	9048608	–	–
nondet.shallow.32	53827061	34531872	81643717	18238494	–	131920587	24814446	29682928	30881951	30881958	17396950	12390833	74123036	69116541
nondet.deep.32	137695422	64288371	200681256	63405414	–	403885291	26178478	42408074	80929142	80928642	43632093	39124284	543614063	539106091

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	mtl.listt	handrolled.listt
countdown.shallow.10000	369.4%	211.1%	600.0%	413.9%	180.6%	217.6%	375.0%	402.8%	244.5%	672.2%	141.7%	100.0%	–	–
countdown.deep.10000	740.6%	285.2%	791.2%	1214.3%	100.0%	120.1%	214.9%	294.6%	181.0%	938.4%	591.8%	591.8%	–	–
catch.shallow.10000	496.7%	161.4%	1402.8%	820.6%	754.0%	26106.7%	341.0%	401.3%	512.2%	512.2%	105.0%	100.0%	–	–
catch.deep.10000	1287.3%	100.0%	2237.2%	2519.1%	467.7%	16201.0%	227.2%	395.1%	317.4%	317.4%	657.9%	701.5%	–	–
local.shallow.10000	643.6%	248.0%	601.8%	330.3%	289.0%	17680.2%	172.2%	199.7%	309.0%	309.0%	100.0%	100.0%	–	–
local.deep.10000	850.3%	143.6%	786.3%	708.7%	167.7%	10256.3%	100.0%	169.6%	179.0%	179.0%	125.6%	225.1%	–	–
nondet.shallow.32	434.4%	278.7%	658.9%	147.2%	–	1064.7%	200.3%	239.6%	249.2%	249.2%	140.4%	100.0%	598.2%	557.8%
nondet.deep.32	526.0%	245.6%	766.6%	242.2%	–	1542.8%	100.0%	162.0%	309.1%	309.1%	166.7%	149.5%	2076.6%	2059.3%